

DiagAccU

Interface Documentation

Version 2.0.0

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Rallou A. Chatzimichail, MEng, MSc, PhD^a, Theodora Chatzimichail, MRCS^b, Aristides T. Hatjimihail, MD, PhD^c

^arc@hcs1.com, ^btc@hcs1.com, ^cath@hcs1.com

Hellenic Complex Systems Laboratory, Kostis Palamas 21, Drama 66131, Greece

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1. Introduction

This guide describes the inputs and controls available in the distributed, compiled DiagAccU 2.0.0 interface. Available inputs depend on the selected plot or table.

The test orientation is $X \geq t$. The population samples, prevalence, and diagnostic threshold are assumed to be mutually independent, with covariance retained between the estimated mean and standard deviation within each population. Calibration uncertainty is specified by the constant coefficient b_0 , the proportional coefficient b_1 , and the selected threshold standard-uncertainty model. It is propagated through a single diagnostic threshold shared by both populations.

2. Interface Controls

2.1. Tabbed Navigation

Choose a primary tab and then the required submodule. Each terminal view has its own controls and saved defaults; changing a control in one view does not synchronize every other view.

2.2. Numerical Settings: Sliders

The program offers controls for numerical settings, which can be adjusted through sliders. Press the 'alt' or 'opt' key while dragging the mouse for more precise control. For even finer adjustments, also press the 'shift' or 'ctrl' keys.

2.3. Non-Numerical Settings

Buttons and drop-down menus select the measure, distribution family, calibration model, and other settings. Available controls depend on the selected view.

2.4. Display Controls

2.4.1. Plots

Extended and limited settings principally change the horizontal profile domain. Threshold plots span the population means plus or minus three or one standard deviations, respectively, with a nonnegative rounded lower bound. Prevalence profiles use 0.001–0.999 or 0.01–0.2. Profiles of b_0 use 0–0.5 times s_H or 0–0.15 times s_H ; profiles of b_1 use 0–0.1 or 0–0.05. The plotted sample-size domain is generated automatically. These settings do not specify the optimization domain.

2.4.2. Tables

Table views provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

3. Input Variables and Parameters

The following ranges describe interface controls. Diseased and nondiseased populations are denoted D and H, and prevalence is v . Arithmetic means and Bessel-corrected standard deviations are supplied for all population families.

$$t: \max(0.01, \min(m_H - 6s_H, m_D - 6s_D)) - \max(m_H + 6s_H, m_D + 6s_D)$$

$$n_D: 2-10,000$$

$$m_D: 0.1-10,000$$

$$s_D: 0.01-1,000$$

$$n_H: 2-10,000$$

$$m_H: 0.1-10,000$$

$$s_H: 0.01-1,000$$

$$v: 0.001-0.999 \text{ in the views that expose prevalence; other views derive } v = \frac{n_D}{n_D + n_H} \text{ from the sample counts.}$$

$$n_U: 20-10,000; \text{ interval views use this control to assign } v_t = n_U - 1 \text{ to the threshold source.}$$

$$b_0: 0 \leq b_0 \leq s_H$$

$$b_1: 0-0.1$$

$$1 - \alpha: 0.900-0.999$$

n : 20-20,000; in the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

The threshold, both means, both standard deviations, and b_0 must use the same measurement unit; b_1 is dimensionless.

In uncertainty profiles parameterized by total sample size and prevalence, the program uses $n_D = vn$, $n_H = (1 - v)n$. Both resulting stratum sizes must exceed one for the core calculation to be valid. Thus some extreme prevalence-sample-size combinations within the slider ranges can produce unavailable results. Noninteger effective sample sizes can occur during a profile sweep.

4. Input Models and Assumptions

A normal, lognormal, or gamma distribution is selected independently for each population. The supplied means and standard deviations are on the arithmetic scale.

The threshold standard uncertainty is specified by a linear model, $u_t = b_0 + b_1 t$, or the root-sum-of-squares form, $u_t =$

$$\sqrt{b_0^2 + (b_1 t)^2}. \text{ Both describe one shared uncertain threshold. The constant and proportional coefficients are supplied inputs.}$$

The diagnostic threshold is supplied in fixed-threshold views and determined automatically in optimized-threshold views. The threshold slider and plot range do not specify the optimization limits.

Interval views accept the nominal level $1 - \alpha$ and quality control (QC) sample count n_U . Changing n_U changes the assigned threshold-source degrees of freedom and the coverage factor; it does not change the threshold standard uncertainty specified by the coefficients.

Prevalence uncertainty is included in the compiled interface. There is no separate control for excluding this contribution.

In the relative-standard-uncertainty profile against total sample size, the F_1 option is listed but does not produce a result.

5. Notation

5.1. Variables and Parameters

t : diagnostic threshold; also used for the optimal threshold

n : total sample size

v : prevalence

b_0 : constant contribution to measurement uncertainty

b_1 : measurement uncertainty proportionality constant

$1 - \alpha$: confidence level

5.2. Diagnostic Accuracy Measures

Se: sensitivity

Sp: specificity

ACC: overall diagnostic accuracy

PPV: positive predictive value

NPV: negative predictive value

DOR: diagnostic odds ratio

LR^+ : positive likelihood ratio

LR^- : negative likelihood ratio

J : Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa.

F_1 : F_1 score.

MCC: Matthews correlation coefficient

5.3. Uncertainty and Interval Outputs

u_c : combined standard uncertainty

u_s : total sampling standard uncertainty

u_m : measurement standard uncertainty

LCL: lower confidence limit

UCL: upper confidence limit

QC: quality control

6. Citation:

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